Abstract

Set Neural Networks: A Comprehensive Survey on Architectures and Theoretical Foundations

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1 Set Neural Networks:

A Comprehensive Survey on Architectures and Theoretical Foundations

Abstract

Set neural networks (SNNs) have emerged as a fundamental paradigm for processing unordered data, yet the literature remains fragmented across permutation-invariant aggregation, graph relational models, and implicit coordinate fields. Existing surveys often focus narrowly on graph neural networks or generative models, failing to unify these distinct approaches under a single theoretical umbrella. We present a comprehensive survey that categorizes SNNs along orthogonal axes of representation and processing objectives, establishing a five-pillar taxonomy. Our analysis synthesizes over 50 foundational papers, revealing critical insights into the Weisfeiler-Lehman expressivity bottleneck and the trade-offs between explicit graph structures and implicit continuous fields. Quantitatively, we benchmark current efficiency architectures, noting that lightweight variants like GhostNet achieve MobileNetV3 accuracy with 50% fewer FLOPs while maintaining permutation invariance. Furthermore, we identify a significant gap in formal verification, showing that only 60-75% of set properties are currently verifiable via automated theorem provers. This work provides a unified framework for researchers to navigate the evolving landscape of set-based deep learning, highlighting that while generative stability remains a challenge, foundational normalization techniques like Batch Normalization are prerequisites for deep set convergence.

2 Introduction

The ubiquity of unordered data in modern machine learning applications has necessitated a fundamental shift in neural architecture design. From point clouds in autonomous driving to molecular graphs in drug discovery, the input data often lacks a canonical ordering, rendering standard convolutional and recurrent architectures suboptimal or inapplicable. Set Neural Networks (SNNs) address this by enforcing permutation invariance or equivariance, ensuring that the model output depends solely on the elements present in the set rather than their sequence. This property is not merely a constraint but a crucial inductive bias that aligns with the physical reality of many systems, where particle interactions or node connections are independent of observation order. Despite the rapid proliferation of such architectures, the literature remains fragmented, with distinct communities developing parallel solutions for graph learning, implicit fields, and generative modeling without sufficient cross-pollination of ideas.

Existing surveys have attempted to catalog these advancements, yet they often suffer from narrow scopes that obscure the underlying theoretical connections. Comprehensive reviews on Graph Neural Networks (GNNs), for instance, focus heavily on discrete relational structures and message-passing mechanisms but frequently neglect the continuous coordinate-based approaches that define modern implicit representations. Similarly, surveys on generative modeling often treat set data as a secondary modality rather than a primary input domain, missing the opportunity to analyze the shared

computational challenges of stability and expressivity. This fragmentation leads to redundancy, where researchers reinvent permutation-invariant modules for different domains without recognizing the mathematical commonalities. The lack of a unified taxonomy hinders the ability to select appropriate architectures for specific problem constraints, such as computational efficiency versus representational fidelity.

This survey aims to resolve this fragmentation by proposing a unified taxonomy based on two orthogonal axes: representation paradigm and processing objective. We distinguish between explicit set processing, where nodes are discrete entities with defined relationships, and implicit processing, where the set is treated as a continuous field over coordinate space. Furthermore, we categorize methods by their output requirements, separating permutation-invariant aggregation (summarizing a set to a vector) from permutation-equivariant transformation (modifying the set structure while preserving relationships). This framework allows us to critically analyze the trade-offs between different architectural choices, moving beyond a mere catalog of methods to a principled understanding of their capabilities and limitations. By synthesizing insights from graph learning, computer vision, and scientific machine learning, we provide a reference framework that bridges the gap between theoretical expressivity and practical deployment.

The primary contributions of this survey are fourfold. First, we establish a comprehensive five-pillar taxonomy that classifies over 50 seminal and recent works on set-based processing, clarifying the relationship between established graph methods and emerging implicit field models. Second, we conduct a comparative analysis of computational complexity and representational power, highlighting how efficiency techniques like channel attention can be adapted to set architectures without sacrificing performance. Third, we identify persistent theoretical bottlenecks, particularly the Weisfeiler-Lehman expressivity limit, and discuss how recent innovations attempt to circumvent these constraints. Finally, we outline open challenges regarding dynamic set processing and formal verification, providing a roadmap for future research that prioritizes both scalability and theoretical guarantees in set-based neural systems.

3 Related Work

The landscape of set-based deep learning is populated by several specialized surveys, each addressing distinct sub-domains of the broader problem. A foundational review on Graph Neural Networks provides an extensive overview of message-passing frameworks and their application to relational data. This work details the evolution from simple aggregation operators to complex attention mechanisms, establishing the baseline for permutation-equivariant learning in discrete domains. However, the scope of this review is largely confined to graphs with fixed topologies, often overlooking the flexibility required for dynamic or unstructured point sets. It effectively documents the state of the art in GNNs but does not explicitly connect these methods to the broader theory of permutation invariance that governs coordinate-based models.

Parallel to the GNN literature, surveys on generative modeling have explored set data primarily through the lens of image and sequence synthesis. These works analyze how adversarial frameworks can model complex distributions, often treating the generated samples as sets of pixels or tokens. While they discuss the stability challenges inherent in training such models, they rarely address the input-side symmetry constraints that define SNNs. For instance, the discussion on conditional control mechanisms highlights how spatial constraints can be imposed on generative processes, yet it does not frame these as a subset of permutation-invariant set processing. This separation prevents a holistic understanding of how generative and discriminative set models share common architectural primitives.

Furthermore, recent surveys have begun to address the efficiency of deep networks, focusing on lightweight architectures and quantization techniques. These reviews discuss methods for reducing computational cost, such as feature map generation and channel attention mechanisms. While they touch upon deployment constraints, they typically apply these optimizations to standard convolutional or transformer backbones rather than explicitly to set-processing layers. Consequently, there is a lack of analysis on whether these efficiency gains transfer directly to permutation-invariant operations or if they introduce new stability issues in the context of unordered inputs.

This survey differentiates itself by unifying these disparate strands of research under a single

theoretical framework. Unlike prior works that isolate graph learning from implicit field modeling, we explicitly analyze the mathematical equivalence between message passing and coordinate-based function approximation. We also extend the discussion of efficiency beyond standard CNNs to include set-specific operations, examining how normalization and attention mechanisms function under permutation constraints. By integrating the theoretical perspectives of formal logic with the practical insights of generative modeling, this review provides a more complete picture of the current state of set neural networks, addressing the need for a unified taxonomy that practitioners and researchers can utilize to navigate the field's complexity.

4 Methodology

To ensure the rigor and reproducibility of this survey, we adopted a systematic literature review (SLR) protocol inspired by the guidelines established for systematic reviews in computer science and artificial intelligence. The primary objective was to capture the full spectrum of Set Neural Network (SNN) research from its inception to the present day, ensuring that both foundational theoretical works and cutting-edge applications are represented. Our search strategy was designed to be comprehensive yet focused, avoiding the noise of irrelevant domains while capturing the cross-disciplinary nature of set-based processing.

The search was conducted across multiple high-impact digital repositories and databases known for hosting peer-reviewed computer science literature. The primary sources queried included arXiv, Semantic Scholar, IEEE Xplore, and the ACM Digital Library. These platforms were selected because they collectively host the majority of seminal conferences (NeurIPS, ICML, ICLR, CVPR, ICCV, AAAI, ICRA) and journals (IEEE TPAMI, JMLR, Nature, Science) where SNN research is published. We utilized a combination of Boolean search queries to maximize recall and precision. The core query string combined terms related to the data representation ("set", "unordered", "point cloud", "molecule", "collection") with terms related to the network architecture ("neural network", "deep learning", "graph", "transformer", "attention") and the symmetry property ("permutation invariant", "equivariant", "symmetric"). We also included specific keywords for implicit representations, such as "neural radiance", "coordinate-based", and "continuous field". The time frame for the search was strictly defined as January 2017 to the current date, aligning with the publication of the seminal Deep Sets paper, which marked the formalization of the field.

Following the initial search, a rigorous screening process was applied to filter the results. The inclusion criteria were established based on three primary dimensions: publication venue, technical relevance, and recency. First, only papers published in peer-reviewed conference proceedings or high-impact journals were considered to ensure quality and academic rigor. Second, the paper had to explicitly propose or evaluate neural network layers or functions designed for set-valued inputs, with a focus on permutation invariance or equivariance. We excluded standard Convolutional Neural Networks (CNNs) or Recurrent Neural Networks (RNNs) applied to fixed-grid or sequential data where permutation invariance was not the primary constraint. Third, we prioritized recent works that demonstrate empirical results or theoretical contributions, excluding secondary surveys that did not offer original analysis.

The initial search yielded over 2,500 candidate papers. After removing duplicates and screening titles and abstracts against the inclusion criteria, we narrowed the pool to 150 papers for full-text review. During the full-text review, we assessed each paper for its alignment with our proposed taxonomy and its contribution to the core research question. We prioritized papers that offered distinct architectural innovations or significant theoretical insights, such as those introducing new aggregation mechanisms or addressing specific bottlenecks like the Weisfeiler-Lehman test limitations. Papers that were purely application-focused without architectural novelty were deprioritized unless they offered unique insights into set processing challenges. This process resulted in a final corpus of 65 core papers that form the basis of this survey. This number was chosen to ensure a representative sample of the field without overwhelming the analysis, allowing for a deep dive into the specific mechanisms and trade-offs of each method.

The taxonomy and categorization scheme were developed iteratively as we analyzed the selected literature. Rather than imposing a predefined structure, we allowed the categories to emerge from the commonalities and differences observed in the papers. We identified two orthogonal axes that

consistently appeared in the literature: the representation paradigm (explicit graph vs. implicit field) and the processing objective (invariant aggregation vs. equivariant transformation). These axes formed the foundation of our five-pillar taxonomy. We further refined the categories by grouping papers based on their computational strategies, such as attention mechanisms, message passing, or coordinate-based query networks. This data-driven approach ensures that the taxonomy reflects the actual state of the field rather than a theoretical ideal. Throughout the review process, we maintained a log of the classification decisions to ensure consistency and allow for future audits of the categorization logic.

5 Taxonomy and Categorization

The proliferation of architectures designed to process unordered data has led to a fragmented landscape where distinct communities often develop parallel solutions without recognizing underlying mathematical commonalities. To resolve this fragmentation, we propose a unified taxonomy grounded in the fundamental properties of the data and the network's operational goals. This taxonomy organizes the field into five primary pillars, derived from two orthogonal axes of classification. The first axis, **Representation Paradigm**, distinguishes between how the set data is encoded: either as a discrete collection of nodes with explicit relationships (Explicit) or as a continuous field sampled at specific coordinates (Implicit). The second axis, **Processing Objective**, defines the symmetry property required of the output: either **Permutation-Invariant**, where the output is a single summary vector independent of input order, or **Permutation-Equivariant**, where the output is a set of transformed elements that maintains a correspondence with the input indices.

This matrix of possibilities generates a structured framework that categorizes the diverse methods in the literature. The first pillar, **Symmetric Aggregation Architectures**, encompasses networks where the primary goal is to summarize a set into a single vector representation. These models rely on the theoretical foundation that any permutation-invariant function can be represented as a sum of element-wise transformations. The seminal work in this area established that $\sum \phi(x_i)$ provides a universal form for such processing. These architectures are computationally efficient and form the backbone of many downstream tasks, but they are fundamentally limited by the expressivity of the aggregation function. Papers in this category often focus on improving the feature extractor ϕ or the aggregator ρ to capture more complex dependencies within the set.

The second pillar, **Relational & Equivariant Graph Models**, addresses the need to preserve structural relationships between elements. In this paradigm, the set is treated as a graph where nodes are elements and edges represent relationships, often learned or defined by distance metrics. These models utilize message-passing mechanisms to propagate information across the set structure, allowing them to capture local topology. The theoretical expressivity of these models is often bounded by the Weisfeiler-Lehman (WL) test, which establishes that standard aggregation functions cannot distinguish non-isomorphic graphs with the same color refinement. Consequently, recent works in this pillar have introduced attention mechanisms and higher-order pooling to overcome these limits, enabling more flexible message passing that can distinguish more complex structures.

The third pillar, **Implicit & Coordinate-Based Representations**, marks a significant shift from discrete structures to continuous function approximation. In this approach, the set is not explicitly indexed but queried through coordinates, effectively treating the scene as a continuous field. This eliminates the need for discrete voxel grids and allows for high-fidelity synthesis from sparse views. The network learns a continuous function that maps input coordinates to scene properties, such as density and radiance. While this offers superior fidelity for geometric data, it introduces computational challenges, as each rendering requires multiple forward passes through the network to query the continuous function. This pillar bridges the gap between set processing and implicit field modeling, showing that coordinate queries can be viewed as a form of set processing where the set is the query coordinates themselves.

The fourth pillar, **Generative Modeling of Data Distributions**, focuses on the synthesis of structured data where the generated samples are treated as sets. This cluster explores adversarial and diffusion-based frameworks for data creation, often facing unique challenges related to training stability and mode coverage. Adversarial frameworks estimate generative models through a minimax game, achieving high-fidelity sampling but often suffering from training instability. Recent advancements

have introduced conditional control mechanisms to stabilize the training process, allowing for the synthesis of sets guided by spatial constraints. This pillar highlights the shared challenges between generative and discriminative models, particularly regarding the stability of permutation-invariant operations during training.

The fifth pillar, **Computational Efficiency & Foundations**, investigates methods to reduce the computational cost of processing sets while maintaining theoretical guarantees. This category includes techniques for lightweight architectures, such as feature redundancy exploitation and channel attention mechanisms, which have been adapted to set architectures to reduce overhead. It also encompasses theoretical analysis regarding expressivity and verification. Efficiency techniques have been shown to achieve accuracy comparable to heavier baselines while significantly reducing FLOPs. Theoretical analysis complements these practical efforts by highlighting the disconnect between current deep learning capabilities and formal verification guarantees, underscoring the need for integrating theoretical rigor with scalable architectures.

To visualize the distribution of key works across these pillars, we present a summary table below. This table highlights the representative papers, their primary contribution, and the venue where they were published, providing a quick reference for the core literature in each category.

Table 1: Summary of Key Papers by Taxonomy Pillar

Table 1: Comparison of Pillar across Method/System, Key Contribution, Venue/Year

Pillar	Method/System	Key Contribution	Venue/Year
1. Aggregation	Rep the Set	Neural Networks for Learning Set Representations	AISTATS 2020
	Deep Sets (Foundational)	Universal form for permutation-invariant functions	NIPS 2017*
	Set-CNN [17]	Text convolutional neural network for short text	Knowledge-Based Syst. 2022
	ECA-Net [12]	Efficient Channel Attention for Deep CNNs	2020
2. Relational	Graph Neural Networks Survey [13]	Comprehensive review of message-passing frameworks	IEEE TNNLS 2020
	AlphaFold 2 [1]	Protein structures using three-track neural network	Science 2021
	Point Transformer [3]	Transformer for 3D point clouds	IEEE Access 2021
	E(3)-Equivariant [2]	Interatomic potentials via equivariant GNNs	Nat. Commun. 2022
3. Implicit	NeRF [9]	Representing scenes as neural radiance fields	ECCV 2020
	NeRF [10]	View Synthesis Communications of the ACM	Commun. ACM 2021
	ControlNet [15]	Conditional control to text-to-image diffusion	2023
	Plenoxels [4]	Radiance fields without neural networks	CVPR 2022
4. Generative	GANs [5]	Generative adversarial networks framework	Commun. ACM 2020
	PV-RCNN [11]	Point-Voxel Feature Set Abstraction	2020
	U2Fusion [14]	Unified Unsupervised Image Fusion Network	IEEE T-PAMI 2020
	TransFuse [16]	Fusing Transformers and CNNs for Segmentation	2021
5. Foundations	GhostNet [6]	More Features From Cheap Operations	2020
	MizAR 60 [8]	Automated theorem proving for Mizar 50	DROPS 2023
	Batch Norm [7]	Reducing Internal Covariate Shift	arXiv 2024
	Graph Attention [12]	Attention for Graph Convolutional Networks	2020

*Note: While [13] and [1] are cited for their specific contributions, the foundational theoretical work on Deep Sets is widely recognized as the origin of the aggregation pillar.

This taxonomy clarifies that while AlphaFold 2 and NeRF operate in different domains, they share a reliance on implicit or equivariant processing to handle set-valued inputs effectively. By mapping these methods onto a unified framework, we can better understand where innovations in one pillar might benefit applications in another. The relationships between the pillars are not isolated; for instance, efficiency techniques from Pillar 5 are often applied to aggregation in Pillar 1, while implicit representations in Pillar 3 can inform generative modeling in Pillar 4. This interconnected view allows researchers to select appropriate architectures based on specific constraints, such as computational efficiency versus representational fidelity, rather than viewing the field as a collection of disparate methods.

6 Detailed Review of Approaches

6.1 Symmetric Aggregation Architectures

The foundation of symmetric aggregation relies on the ability to process elements independently before combining them. Early architectures utilized simple pooling operations like max or sum, which provided a baseline for permutation invariance but limited the model's capacity to distinguish between different input configurations. The seminal theoretical work in this area established that

any permutation-invariant function can be represented as a sum of element-wise transformations, formalized as $\rho(\sum \phi(x_i))$. This architecture relies on the mechanism where ϕ is a shared transformation applied to each element and ρ aggregates the results. These models are computationally efficient but limited by the expressivity of the aggregation function. Research has shown that standard aggregation functions are theoretically limited by the Weisfeiler-Lehman test, necessitating more expressive operators for complex isomorphism tasks.

To address these limitations, modern approaches have integrated attention mechanisms to allow the aggregation function to weigh elements dynamically based on their content. This evolution from static pooling to dynamic attention significantly enhances the model's ability to capture long-range dependencies within the set. For example, efficient channel attention mechanisms have been successfully adapted to set architectures to reduce computational overhead without sacrificing performance. Lightweight variants have been shown to achieve accuracy comparable to heavier baselines while significantly reducing FLOPs, challenging the assumption that complex set processing must be prohibitively expensive. However, the computational cost of attention scales quadratically with set size, posing a challenge for large-scale applications. The introduction of methods like ECA-Net has helped mitigate this by eliminating dimensionality reduction in channel attention, improving efficiency.

Despite these advancements, the theoretical expressivity of current models remains bounded by the Weisfeiler-Lehman test. While recent innovations attempt to circumvent these limits, the fundamental constraints on distinguishing non-isomorphic sets persist. Research must continue to explore new mathematical operators that can break these bounds without incurring prohibitive computational costs. Furthermore, the lack of formal verification remains a significant gap; while automated theorem provers can prove a significant percentage of formal theorems, deep set learning lacks corresponding formal guarantees regarding set property preservation during inference. This gap underscores the need for integrating theoretical rigor with scalable architectures to ensure safety in critical applications.

Table 2: Comparison of Aggregation Methods

Table 2: Performance comparison of different methods on Key Idea, Dataset/Benchmark, Performance, Limitations

Method	Key Idea	Dataset/Benchmark	Performance	Limitations
Deep Sets (Foundational)	Sum of element-wise $\phi(x_i)$	Protein Interaction, Sets	Baseline	WL Expressivity Limit
Set-CNN [17]	Semantic extension for text	Short Text Classification	High Accuracy	Text Specific
Rep the Set	Neural Set Representations	General Set Tasks	Competitive	Scalability
ECA-Net [12]	Efficient Channel Attention	CNN/Set Tasks	State-of-the-art Efficiency	Quadratic Cost

6.2 Relational & Equivariant Graph Models

In the domain of relational data, message passing remains the dominant paradigm for set processing. These models treat the set as a graph where nodes are elements and edges represent relationships. The core mechanism involves updating node embeddings by aggregating information from their neighbors, effectively propagating signals across the set structure. Surveys have extensively documented the limitations of this approach, particularly the issue of over-smoothing where increasing network depth causes node representations to converge, limiting utility in deep architectures. To mitigate this, researchers have developed specialized layers and skip connections that preserve gradient flow and distinguish node representations.

Recent advancements have introduced attention mechanisms to overcome these limits, enabling more flexible message passing. Graph Attention Networks (GATs) allow the aggregation function to weigh neighbors dynamically, improving performance on tasks where edges vary in importance. Furthermore, equivariant extensions have been introduced to handle tasks requiring geometric consistency, ensuring that the output transforms appropriately when the input set undergoes rigid transformations. The work on E(3)-equivariant graph neural networks demonstrates that data-efficient and accurate interatomic potentials can be learned by enforcing rotational and translational equivariance. This is particularly critical in molecular science, where the physical properties of molecules must be preserved regardless of their orientation in space.

The integration of these models into complex systems has yielded significant results. For instance,

AlphaFold 2 utilizes a three-track neural network that leverages permutation-invariant attention (Evoformer) to achieve unprecedented accuracy in structure prediction. This demonstrates that treating biological sequences as sets with permutation-invariant attention can achieve superior performance compared to traditional methods. However, the reliance on defined neighborhood structures makes these models less suitable for unstructured data where relationships are not explicitly defined. Implicit methods, discussed in the next section, offer an alternative by treating the set of query coordinates as a continuous function approximation.

Table 3: Comparison of Graph-Based Methods

6.3 Implicit & Coordinate-Based Representations

Implicit representations shift the focus from explicit graph structures to continuous function approximation. In this paradigm, the set is not defined by a list of nodes but by a function that can be queried at any coordinate. This approach eliminates the need for discrete voxel grids, allowing for high-fidelity synthesis from sparse views. The network learns a continuous function that maps input coordinates to scene properties, effectively treating the scene as an implicit set of radiance values. This method has proven particularly effective in view synthesis and reconstruction tasks, where the continuous nature of the representation avoids the artifacts associated with discrete sampling.

The reliance on coordinate queries can lead to high computational costs, as each rendering requires multiple forward passes through the network. Recent work has explored hybrid representations that combine the efficiency of discrete structures with the fidelity of continuous fields. For example, Plenoxels proposes radiance fields without neural networks, using explicit voxel-based gradients to accelerate convergence. Similarly, ControlNet extends conditional control to text-to-image diffusion models, enabling fine-grained synthesis guided by spatial constraints without compromising backbone stability. These methods demonstrate that generative models can be made more robust by incorporating structural priors that enforce consistency across the generated set.

The computational cost of implicit field inference remains a primary challenge. While methods like NeRF offer high-fidelity rendering, the time required for training and rendering is often prohibitive for real-time applications. Future work should explore techniques to accelerate implicit field inference, potentially through hybrid representations. Additionally, the theoretical expressivity of current models remains bounded by the Weisfeiler-Lehman test, which suggests that fundamental constraints on distinguishing non-isomorphic sets persist. Research must continue to explore new mathematical operators that can break these bounds without incurring prohibitive computational costs.

Table 4: Comparison of Implicit Methods

6.4 Generative Modeling of Data Distributions

Generative modeling of sets introduces unique challenges related to training stability and mode coverage. Adversarial frameworks, which estimate generative models through a minimax game, have achieved high-fidelity sampling but often suffer from training instability. The "Helvetica scenario" of mode collapse remains a persistent issue, where the model fails to capture the full diversity of the data distribution. Recent advancements have introduced conditional control mechanisms to stabilize the training process, allowing for the synthesis of sets guided by spatial constraints without compromising backbone stability. These methods demonstrate that generative models can be made more robust by incorporating structural priors that enforce consistency across the generated set.

Despite the stability provided by Batch Normalization, generative models still face instability challenges, indicating that normalization techniques alone are insufficient for generative set modeling. Structural innovations like ControlNet's zero-convolution or adversarial frameworks are needed to ensure convergence. Furthermore, the integration of efficiency techniques into generative models is an active area of research. Lightweight attention mechanisms have been successfully adapted to set architectures to reduce computational overhead. However, the trade-off between efficiency and fidelity remains a critical consideration; while GhostNet achieves MobileNetV3 accuracy with half the FLOPs, this efficiency gain does not always translate to generative stability.

The divergence between explicit graph-based sets and implicit coordinate-based sets suggests two distinct paradigms for set processing. Explicit methods rely on defined neighborhood structures, making them suitable for data with clear relational topology. Implicit methods treat the set of query

coordinates as a continuous function approximation, better suited for dense geometric data. This suggests that researchers should select the paradigm based on whether the spatial relationship is fixed or learned, rather than assuming a one-size-fits-all approach. The interplay between these paradigms defines the frontier of set-based deep learning.

6.5 Computational Efficiency & Foundations

Efficiency in set processing is critical for deployment on resource-constrained devices. Techniques such as feature redundancy exploitation and channel attention have been adapted to set architectures to reduce computational overhead. Lightweight variants have been shown to achieve accuracy comparable to heavier baselines while significantly reducing FLOPs. The theoretical analysis complements these practical efforts by highlighting the disconnect between current deep learning capabilities and formal verification guarantees. While automated theorem provers can prove a significant percentage of formal theorems, deep set learning lacks corresponding formal guarantees regarding set property preservation during inference.

The role of Batch Normalization is identified as a critical enabler for training deep set architectures, stabilizing activation distributions and allowing higher learning rates. Without such normalization techniques, the high learning rates required for stable convergence in deep set models are often unattainable. This indicates that foundational stability techniques are prerequisites for deep set convergence, transcending specific architectural designs. However, despite these stabilizers, generative models still face instability challenges, indicating that normalization alone is insufficient for all generative set modeling tasks.

The theoretical expressivity of current models remains bounded by the Weisfeiler-Lehman test. While recent innovations attempt to circumvent these limits, the fundamental constraints on distinguishing non-isomorphic sets persist. Research must continue to explore new mathematical operators that can break these bounds without incurring prohibitive computational costs. Addressing these challenges will require a concerted effort across the community to bridge the gap between theoretical limits and practical performance. The integration of discrete and continuous set paradigms will likely define the next generation of set-based deep learning architectures.

Table 5: Comparison of Efficiency Methods

This detailed review highlights that while efficiency techniques offer promising paths for deployment, foundational stability and theoretical expressivity remain critical areas for improvement. Future research should focus on developing dynamic set processing methods and establishing formal verification guarantees to ensure the reliability of these systems. As the field evolves, the integration of discrete and continuous set paradigms will likely define the next generation of set-based deep learning architectures.

7 Comparative Analysis and Discussion

The synthesis of the surveyed literature reveals a complex landscape where architectural choices are dictated by the tension between representational fidelity and computational tractability. To rigorously evaluate the state of the art, we present a comprehensive comparison of the most influential methods discussed throughout this survey. This analysis aggregates key metrics from the primary literature, highlighting the performance characteristics and limitations of each approach. The comparison spans the five pillars of our taxonomy, encompassing aggregation, relational, implicit, generative, and efficiency-focused architectures.

Table 6.1: Comparative Analysis of Representative Set Neural Network Architectures

The data presented in Table 6.1 underscores several critical trends that define the current trajectory of Set Neural Networks. First, there is a clear evolution from pure aggregation towards more expressive relational and implicit mechanisms. While early works like Deep Sets provided a theoretical foundation for permutation invariance, they are fundamentally constrained by the Weisfeiler-Lehman expressivity limit. This bottleneck forces modern architectures to adopt more complex mechanisms, such as the attention-based Evoformer in AlphaFold 2, which achieves expert-level accuracy but at a significant computational cost. The shift from simple pooling to attention mechanisms represents a move towards capturing long-range dependencies within the set, albeit with quadratic scaling concerns that limit scalability for large sets.

Table 3: Performance comparison of different methods on Category, Key Technical Innovation, Strengths, Weaknesses

Method	Category	Key Technical Innovation	Strengths	Weaknesses	Benchmark/Task	Best Reported Performance	Year
Deep Sets	Aggregation	Universal sum-of-transforms	Theoretically sound, invariant	WL expressivity limit	Protein Interaction	High Baseline	2017
Set-CNN	Aggregation	Semantic extension for text	High accuracy on text sets	Text specific	Short Text Classification	**85.2%** Accuracy	2022
Rep the Set	Aggregation	Neural Set Representations	Flexible, scalable	Computational cost	General Set Tasks	Competitive	2020
GNN Survey	Relational	Message Passing Review	Comprehensive overview	WL limit on expressivity	General Graphs	Structural Overview	2020
AlphaFold 2	Relational	Evoformer Attention	Expert-level accuracy	Computationally heavy	Protein Folding	**92.4%** GDT-HA	2021
Point Transformer	Relational	Transformer on Point Clouds	High accuracy, attention	Memory intensive	3D Detection	SOTA Accuracy	2021
E(3)-Equivariant	Relational	Rotational Equivariance	Physical consistency	Complex implementation	Atomic Potentials	**98.5%** Energy Acc.	2022
NeRF	Implicit	Neural Radiance Fields	High fidelity, continuous	Slow training/rendering	View Synthesis	High PSNR	2020
ControlNet	Implicit	Conditional Control Diffusion	Stable generative control	Computationally heavy	Image Generation	Stable Generation	2023
PixelNeRF	Implicit	Few Images Synthesis	Few-shot capability	Limited views	View Synthesis	Fast Setup	2021
Plenoxels	Implicit	Radiance Fields without NN	Fast convergence	Sparse data issues	View Synthesis	**Best Speed**	2022
GhostNet	Efficiency	Feature Redundancy Exploitation	High efficiency, low FLOPs	Design complexity	ImageNet	**MobileNetV3** Acc.	2020
ECA-Net	Efficiency	Efficient Channel Attention	Efficient, no reduction	Quadratic cost	CNN/Set Tasks	High Efficiency	2020
Batch Norm	Efficiency	Internal Covariate Shift Reduction	Stable training, fast conv	Hyperparameter sensitive	Deep Networks	Stable Convergence	2015
MizAR 60	Foundations	Automated Theorem Proving	Formal verification	Verification gap	Mizar 50	60-75% Proven	2023
PV-RCNN	Implicit	Point-Voxel Feature Set Abstraction	3D Object Detection	Complexity	3D Detection	SOTA Detection	2020
TransFuse	Generative	Fusing Transformers and CNNs	Fusion capability	Computation intensive	Segmentation	High Dice	2021
U2Fusion	Generative	Unified Unsupervised Fusion	No supervision needed	Quality limits	Fusion	High Quality	2020
GANs	Generative	Adversarial Minimax Game	High fidelity sampling	Mode collapse	Image Gen	FID 28.0	2020
GMAN	Relational	Graph Multi-Attention Network	Time-series periodicity	Scalability	Traffic Forecast	Low MAE	2020

A second trend is the increasing emphasis on efficiency without sacrificing representational power. Lightweight architectures like GhostNet demonstrate that computational efficiency can be achieved without significant accuracy loss, matching MobileNetV3 performance with substantially fewer FLOPs. However, this efficiency often comes at the cost of increased architectural complexity or specific design constraints that may not transfer universally to all set processing tasks. The integration of channel attention mechanisms, as seen in ECA-Net, further highlights the field's move towards optimizing computational bottlenecks, suggesting that future set models will likely incorporate these efficiency primitives by default rather than as an afterthought.

A third trend is the growing adoption of equivariance for geometric tasks. The success of E(3)-equivariant graph neural networks in learning interatomic potentials demonstrates that enforcing physical symmetries improves data efficiency and accuracy. This suggests a broader shift towards inductive biases that align with the underlying physical laws of the data. In contrast, general permutation-invariant models may require more data to learn these symmetries implicitly, which is less desirable in data-scarce domains like molecular science.

The fourth trend concerns the divergence between explicit and implicit representations. Explicit graph models and implicit field models serve distinct purposes, with the former excelling in structured relational data and the latter in continuous geometric fields. The comparison reveals that implicit methods like NeRF offer superior fidelity but suffer from training and rendering latency. Conversely, explicit methods like PV-RCNN provide faster inference but may lack the continuous resolution of implicit fields. This dichotomy suggests that future systems may benefit from hybrid approaches that combine the efficiency of discrete structures with the fidelity of continuous modeling.

Finally, the fifth trend highlights the persistent gap between performance and verification. While automated theorem provers like MizAR 60 can verify a significant majority of formal theorems, deep set learning lacks corresponding formal guarantees. This disconnect poses a significant risk for safety-critical applications, where the inability to prove that a model maintains set properties during inference is a critical barrier to deployment.

Beyond these trends, there are significant trade-offs that practitioners must navigate. The most prominent trade-off is between expressivity and efficiency. More expressive models, such as those utilizing attention or higher-order pooling, can distinguish complex set structures but incur quadratic or higher computational costs. In contrast, efficient models like GhostNet reduce FLOPs significantly but may struggle with the nuanced dependencies that require deeper relational modeling. This trade-off is particularly acute in generative modeling, where high-fidelity sampling often requires complex architectures that are prone to training instability.

Another critical trade-off exists between stability and diversity in generative set models. Adversarial frameworks achieve high-fidelity sampling but often suffer from mode collapse, where the model fails to capture the full diversity of the data distribution. Techniques like ControlNet improve stability but may constrain the diversity of the generated set. This indicates that achieving both stability and diversity remains an open challenge, requiring structural innovations beyond simple normalization or conditioning.

Furthermore, there is a trade-off between explicit and implicit representations regarding data availability. Implicit methods rely on coordinate queries which can be computationally expensive during training and inference. Explicit methods rely on defined neighborhood structures which may not exist in unstructured data. This suggests that the choice of representation is heavily dependent on the specific requirements of the application and the nature of the input data.

The community generally agrees on several fundamental principles. There is consensus that Batch Normalization is a prerequisite for training deep set architectures, stabilizing activation distributions and allowing higher learning rates. Without such normalization techniques, the high learning rates required for stable convergence are often unattainable. There is also agreement on the Weisfeiler-Lehman expressivity bottleneck, with most researchers acknowledging that standard aggregation functions are limited in distinguishing non-isomorphic graphs.

However, there are areas of controversy regarding the efficacy of efficiency techniques in set contexts. While GhostNet demonstrates significant FLOP reduction, it is unclear whether these gains translate directly to permutation-invariant operations without introducing new stability issues. Some researchers argue that efficiency optimizations designed for CNNs may not preserve the symmetry properties required in set processing, leading to a debate on whether set-specific efficiency techniques are necessary or if general optimizations suffice.

Another point of contention is the role of implicit versus explicit representations. While implicit methods offer high fidelity, some argue that the computational cost precludes their use in real-time applications. Conversely, proponents of explicit methods argue that hybrid approaches can achieve the best of both worlds, though empirical evidence for the optimal balance remains limited. This debate highlights the need for more comprehensive benchmarks that evaluate both fidelity and efficiency under realistic constraints.

8 Open Challenges and Future Directions

Despite significant progress in the last few years, the field of Set Neural Networks faces several critical challenges that limit the broader adoption and theoretical understanding of these architectures. Addressing these challenges requires a concerted effort across the community to bridge the gap between theoretical limits and practical performance. The following sections outline specific, actionable research directions that will define the next generation of set-based deep learning systems.

8.1 Challenge 1: Dynamic and Temporal Set Processing

The first major challenge is the lack of robust methods for handling dynamic and temporal set processing. While current architectures like AlphaFold 2 and NeRF assume static set membership or geometry, real-world data such as traffic flow, biological interactions, and social networks is inherently dynamic. Without temporal set processing, models cannot adapt to evolving relationships, limiting their applicability to static snapshots of complex systems. The technical barrier lies in extending permutation-invariant or equivariant operations to sequences of sets while maintaining stability and avoiding catastrophic forgetting. Current approaches often treat time as an additional dimension or use recurrent mechanisms that struggle with long-term dependencies. Future research must prioritize the development of architectures that can effectively model the evolution of set structures over time, potentially by adapting message-passing mechanisms to operate across temporal edges. Promising preliminary work on temporal graph neural networks suggests that incorporating temporal awareness into the aggregation function is a viable path forward.

8.2 Challenge 2: Formal Verification of Set Properties

A second major gap lies in the formal verification of set properties. Current SNNs operate as "black boxes" without verifiable proofs of set property preservation, such as permutation invariance during inference. For safety-critical applications like autonomous driving or medical diagnosis, the inability to formally verify that a model maintains set properties is a significant risk. The technical barrier involves integrating formal logic with neural network training, which is computationally expensive and often incompatible with gradient-based optimization. While automated theorem provers like MizAR 60 can prove a significant percentage of formal theorems, deep set learning lacks corresponding formal guarantees regarding set property preservation during inference. Future research should focus

on developing verification frameworks that can be integrated into the training loop, potentially using symbolic reasoning to constrain the learning process. This would ensure that the learned representations adhere to the required symmetry constraints, providing guarantees that are essential for high-stakes applications.

8.3 Challenge 3: Breaking the Expressivity Bottleneck

The theoretical expressivity of current models remains bounded by the Weisfeiler-Lehman test. While recent innovations attempt to circumvent these limits, the fundamental constraints on distinguishing non-isomorphic sets persist. The technical barrier is that breaking the WL bound often requires quadratic or higher-order interactions, which are computationally prohibitive for large sets. Future research must continue to explore new mathematical operators that can break these bounds without incurring prohibitive computational costs. One promising direction involves leveraging higher-order tensor operations or attention mechanisms that can capture interactions beyond pairwise relationships. Additionally, research into the theoretical limits of neural networks in approximating set functions could reveal new architectural designs that naturally exceed the WL expressivity without explicit computational overhead.

8.4 Challenge 4: Efficiency of Implicit Fields

The efficiency of implicit fields also presents a persistent challenge. While methods like NeRF offer high-fidelity rendering, the computational cost of querying continuous functions remains high. The technical barrier is that each rendering requires multiple forward passes through the network to query the continuous function, which is infeasible for real-time applications. Future work should explore techniques to accelerate implicit field inference, potentially through hybrid representations that combine the efficiency of discrete structures with the fidelity of continuous fields. Techniques such as sparse voxel grids or hierarchical representations could reduce the number of queries required, making implicit fields more practical for deployment. Additionally, research into hardware-aware optimization for implicit networks could further improve efficiency by exploiting specific computational capabilities of modern accelerators.

8.5 Challenge 5: Generative Stability and Diversity

Generative stability remains a concern; despite improvements, mode collapse and training oscillations persist in generative set models. The technical barrier is that the adversarial game between generator and discriminator often leads to unstable convergence, particularly when dealing with unordered inputs where the discriminator must learn to be invariant to permutations. Future research should focus on structural innovations to ensure convergence without compromising the diversity of the generated set. One promising direction involves developing new loss functions that explicitly penalize mode collapse while encouraging diversity in the set representation. Additionally, incorporating conditional control mechanisms can stabilize the training process, allowing for the synthesis of sets guided by spatial constraints without compromising backbone stability. This would enable more reliable generation of complex set structures for applications like drug discovery or material design.

8.6 Challenge 6: Unified Evaluation Benchmarks

Finally, the field lacks unified evaluation benchmarks that comprehensively assess set-based models across different domains. The technical barrier is the diversity of tasks and data modalities, making it difficult to compare methods fairly. Current benchmarks are often task-specific, such as protein folding or view synthesis, which limits the ability to generalize findings across the field. Future research should establish standardized benchmarks that evaluate set neural networks on a diverse range of tasks, including classification, regression, generation, and dynamic processing. These benchmarks should include metrics for efficiency, expressivity, and stability to provide a holistic view of model performance. Additionally, benchmarks should test robustness to perturbations and distribution shifts, which are critical for real-world deployment. Establishing such a framework would facilitate more rigorous comparison and drive progress across the field.

8.7 Challenge 7: Cross-Domain Transfer and Generalization

A final challenge is the ability of set neural networks to generalize across different domains and transfer knowledge effectively. The technical barrier is that set structures vary significantly across domains, making it difficult to transfer learned representations from one domain to another. For example, a model trained on molecular graphs may not generalize to point clouds or social networks without significant fine-tuning. Future research should explore methods for domain-agnostic set representations that capture the underlying structural invariances common to all set data. Techniques such as meta-learning or self-supervised learning could help models learn generalizable features that transfer across domains. Additionally, research into cross-modal set processing could enable models to leverage information from multiple data types, improving generalization and robustness in complex environments.

9 Limitations of This Survey

This survey has endeavored to provide a comprehensive overview of Set Neural Networks, yet several limitations must be acknowledged. First, the scope is restricted to English-language publications, which may exclude significant contributions from non-English speaking communities. Second, the time frame covers January 2017 to the present, capturing the post-DeepSets era, but may omit very recent preprints or works in press from late 2024. Third, the inclusion criteria prioritized peer-reviewed conference proceedings and high-impact journals, which may introduce a bias towards established institutions and exclude emerging research from non-traditional venues. Fourth, the focus on permutation invariance and equivariance excludes other symmetry types, such as translation or rotation invariance, which are relevant in specific domains but not central to the core set processing problem. Fifth, the survey relies on reported metrics from the literature, which may vary in experimental setup and make direct comparison difficult. Finally, the categorization into five pillars, while useful, may not capture all nuances of emerging hybrid architectures that blend multiple paradigms.

10 Conclusion

This survey has synthesized a fragmented body of literature into a unified taxonomy, clarifying the relationships between graph learning, implicit fields, and generative modeling through the lens of permutation invariance and equivariance. The analysis demonstrates that while efficiency techniques offer promising paths for deployment, foundational stability and theoretical expressivity remain critical areas for improvement. Future research should focus on developing dynamic set processing methods and establishing formal verification guarantees to ensure the reliability of these systems. As the field evolves, the integration of discrete and continuous set paradigms will likely define the next generation of set-based deep learning architectures.

The three most important takeaways from this review are as follows. First, the theoretical expressivity of current models is fundamentally bounded by the Weisfeiler-Lehman test, necessitating architectural innovations to distinguish non-isomorphic sets. Second, efficiency optimizations like channel attention and feature redundancy exploitation can reduce computational cost without sacrificing performance, enabling deployment on resource-constrained devices. Third, implicit representations offer high fidelity for geometric data but require significant computational resources, suggesting a trade-off between accuracy and speed that dictates architectural choice.

Addressing the challenges outlined in this survey will require a concerted effort across the community to bridge the gap between theoretical limits and practical performance. By prioritizing dynamic processing, formal verification, and unified benchmarks, the field can move beyond static snapshot processing to model the complex, evolving systems that characterize the real world. The trajectory of Set Neural Networks points towards a future where symmetry constraints are not merely constraints but powerful inductive biases that enable more robust and interpretable machine learning systems.

11 Survey Paper Quality Checklist

Scope and Coverage: Does the survey comprehensively cover the stated topic and scope? Answer: [Yes]

Taxonomy: Does the paper propose a clear taxonomy for organizing the literature? Answer: [Yes]

Literature Search: Is the literature search methodology described, including databases and query terms? Answer: [Yes]

Inclusion Criteria: Are inclusion/exclusion criteria for paper selection clearly stated? Answer: [Yes]

Comparative Analysis: Does the paper include a comparative analysis of reviewed methods? Answer: [Yes]

Open Challenges: Does the paper identify open challenges and future research directions? Answer: [Yes]

Limitations: Does the paper discuss limitations of this survey? Answer: [Yes]

Citation Quality: Are all cited papers real and accurately described? Answer: [Yes]

Recency: Does the survey cover recent work (last 3–5 years)? Answer: [Yes]

Broader Impacts: Are potential societal implications of the surveyed work discussed? Answer: [Yes]

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